ONE-PAGE BRIEF

Post-quantum cryptography review: reddsa @ 04e973bcf8d4

Client: Zcash Foundation — public repository (passive review) · Source: github.com/ZcashFoundation/reddsa, commit `04e973bcf8d4` · Reviewed: 2026-11-04 19:15 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 0 of 4

no post-quantum cryptography

FINDINGS

0 Critical **0** High **2** Medium

0 Low **2** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium — QSI-2026-12470

Quantum-vulnerable FROST: threshold signing and MPC protocols (1 file(s)) preliminary

What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...

Medium — QSI-2026-12469

Quantum-vulnerable RedDSA: role not determined (3 file(s)) preliminary

What to do: Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an abstraction layer for...

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium threshold signing and MPC protocols	FROST	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the...
Medium role not determined	RedDSA	Classify each usage, record data lifetimes, and plan migration to a post-quantum algorithm; introduce an...

Static review of source code and manifests; files analysed: 15. No requests were sent to the owner's systems.

Full report, content SHA-256: `5ca6c9c5ed7313b21cef308aaed72429129f081cd89916ded12aeb8640a00654`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.